

Meissner Minimality in the Confocal Peabody Family

Bruce Swihart*

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Abstract

Arelio, Montejano, and Oliveros introduced a three-parameter family of bodies of constant width obtained from a regular Reuleaux tetrahedron by replacing its six edge neighborhoods with envelopes of spheres associated with confocal quadrics. The family contains the two classical Meissner bodies as a circle-line degeneration and Robert's tetrahedrally symmetric body as a parabolic limit. We prove that the Meissner degenerations minimize volume within this regular-tetrahedron confocal family. We first verify that the explicit eccentricity chart satisfies the source construction for every parameter and every orientation, and we prove the almost-everywhere partition of the normal sphere used in the area bookkeeping. This gives the exact decomposition

$$\text{Vol } K(\mathbf{e}, \sigma) = V_M + \Phi(e_1) + \Phi(e_2) + \Phi(e_3),$$

independently of the eight orientation choices. A complete one-pair calculation reduces Φ to a fixed integral with a globally controlled principal-arctangent branch. The stable chart extends real analytically to the parabolic endpoint. An independent directed-rounding MPFR certificate proves the high-headroom inequalities

$$\Phi(1) > \frac{1}{4000}, \quad \Phi''(e) < -\frac{1}{30000} \quad (0 < e < 1).$$

Concavity then gives $\Phi(e) > e/4000$. Equality holds exactly when all three parameters vanish. The result is restricted to regular-tetrahedron confocal Peabodies and does not resolve the three-dimensional Blaschke–Lebesgue problem.

Keywords: bodies of constant width, Peabodies, Meissner bodies, confocal quadrics, strict concavity, MPFR, computer-assisted proof

1 Introduction

The Blaschke–Lebesgue problem asks for convex bodies of minimum volume among all convex bodies of prescribed constant width. In the plane, the minimizers are the Reuleaux triangles [3, 6]. The corresponding three-dimensional problem remains open. Bonnesen and Fenchel conjectured that the two noncongruent Meissner bodies are minimizers; their common width-one volume is

$$V_M = \pi \left(\frac{2}{3} - \frac{\sqrt{3}}{4} \arccos \frac{1}{3} \right) = 0.419860045965080 \dots \quad (1.1)$$

See [5, 7, 8] for background.

*DCG major-revision checkpoint. The official-header MPFR replay and independent geometric proofreading remain pre-submission gates.

Arelio, Montejano, and Oliveros introduced *Peabodies*, obtained by replacing neighborhoods of the six edges of a Reuleaux tetrahedron with sections of envelopes of spheres guided by confocal quadrics [1]. Their regular-tetrahedron construction gives genuine bodies of constant width and contains both Meissner bodies and Robert's tetrahedrally symmetric body. We study only this regular-tetrahedron family.

The regular tetrahedron has three pairs of opposite edges. Each pair has a parameter $e_i \in [0, 1)$, and an orientation bit $\sigma_i \in \{0, 1\}$ records which edge receives the elliptic device. Let

$$\mathbf{e} = (e_1, e_2, e_3), \quad \boldsymbol{\sigma} = (\sigma_1, \sigma_2, \sigma_3),$$

and write $K_d(\mathbf{e}, \boldsymbol{\sigma})$ for the resulting body of width d . The value $e_i = 0$ is the circle-line Meissner degeneration; the limit $e_i \rightarrow 1^-$ is parabolic.

Theorem 1.1 (Main theorem). *For every $d > 0$, every $\mathbf{e} \in [0, 1)^3$, and every $\boldsymbol{\sigma} \in \{0, 1\}^3$,*

$$\text{Vol } K_d(\mathbf{e}, \boldsymbol{\sigma}) \geq \pi \left(\frac{2}{3} - \frac{\sqrt{3}}{4} \arccos \frac{1}{3} \right) d^3. \quad (1.2)$$

Equality holds if and only if $e_1 = e_2 = e_3 = 0$. Over the eight orientation choices, the equality cases form exactly the two classical Meissner congruence classes.

The proof has two parts. First, the source geometry and a partition of the unit normal sphere give an exact additive volume formula. Second, a one-pair calculation and a compact directed-rounding certificate prove a positive parabolic endpoint and uniform strict concavity. Positivity then follows from the chord inequality.

2 Full-parameter Peabody geometry and exact additivity

We work first at the native scale of the source construction: a regular tetrahedron of side length two and a body of constant width two.

2.1 The explicit chart lies in the source family

For one opposite-edge pair, let $0 \leq e < 1$ and put

$$b^2 = \frac{3 + 3e^2 + 4\sqrt{2}e}{1 - e^2}, \quad a = \frac{b}{\sqrt{1 - e^2}}. \quad (2.1)$$

Since $b^2 \geq 3$, define

$$\theta = \arccos \frac{1}{b}, \quad \eta = \text{arsinh} \frac{1}{b}, \quad u_0 = \sin \theta, \quad v_0 = \cosh \eta.$$

In an orthonormal frame $(\mathbf{k}, \mathbf{p}, \mathbf{q})$, set

$$X(t) = a \sin t \mathbf{k} + b \cos t \mathbf{p}, \quad \theta \leq t \leq \pi - \theta, \quad (2.2)$$

$$Y(\xi) = ae \cosh \xi \mathbf{k} + b \sinh \xi \mathbf{q}, \quad -\eta \leq \xi \leq \eta. \quad (2.3)$$

The curves lie in orthogonal planes, have a common axis, and are confocal because

$$a^2 - b^2 = a^2 e^2.$$

Their transverse endpoint coordinates satisfy

$$b \cos \theta = 1, \quad b \sinh \eta = 1,$$

so the corresponding longitudinal beams are the prescribed opposite edges of length two.

Define the pea radii

$$R_E(t) = ae(\sin t - u_0), \quad R_H(\xi) = a(v_0 - \cosh \xi). \quad (2.4)$$

They are nonnegative on the parameter rectangle and vanish at the beam endpoints.

Lemma 2.1 (Full-parameter admissibility). *For every $e \in (0, 1)$, the curves (2.2)–(2.3) with radii (2.4) form a pair of convex confocal pea-pod devices satisfying the hypotheses of [1, Theorem 4.5]. At $e = 0$ the pair is the circle-line degeneration of [1, Section 5.3]. Interchanging the two devices preserves admissibility.*

Proof. Let $B = b^2$. Directly from (2.1),

$$B + 1 = \frac{2(e + \sqrt{2})^2}{1 - e^2}, \quad B - 1 = \frac{2(\sqrt{2}e + 1)^2}{1 - e^2}.$$

All factors are positive, so

$$\sqrt{B + 1} - e\sqrt{B - 1} = 2\sqrt{1 - e^2}.$$

Since $v_0 = \sqrt{B + 1}/b$ and $u_0 = \sqrt{B - 1}/b$, this is

$$a(v_0 - eu_0) = 2. \quad (2.5)$$

A calculation given in Section 3 shows

$$|X(t) - Y(\xi)| = a(\cosh \xi - e \sin t). \quad (2.6)$$

Combining (2.4), (2.5), and (2.6) gives

$$|X - Y| + R_E + R_H = 2. \quad (2.7)$$

Thus the explicit chart is precisely a convex confocal device pair in the sense of [1, Definition 3.5 and Theorem 3.3]. The endpoint statement is [1, Section 5.3]. Swapping the two devices changes only their labels. $\square$

Apply Theorem 2.1 independently to all three opposite-edge pairs. An orientation bit merely exchanges the two devices in one pair. Hence [1, Theorem 4.5] applies for every $\mathbf{e} \in [0, 1]^3$ and every $\boldsymbol{\sigma} \in \{0, 1\}^3$: the six wedge-pod patches and four spherical caps bound a convex body of constant width two.

2.2 Patch decomposition and normal-sphere partition

For generic parameters, denote the open spherical-cap interiors by

$$C_A^\circ, C_B^\circ, C_C^\circ, C_D^\circ$$

and the six open wedge-pod interiors by

$$W_1^{+, \circ}, W_1^{-, \circ}, \dots, W_3^{+, \circ}, W_3^{-, \circ}.$$

The complement of these ten open patches is a finite union of seam curves and the four vertices. The source construction is smooth along nonvertex seams.

For a strictly convex body K , write $R_K(n)$ for the unique support point with outward normal $n \in \mathbb{S}^2$.

Lemma 2.2 (Opposite support points). *For a body of constant width two,*

$$R_K(-n) = R_K(n) - 2n. \quad (2.8)$$

Proof. Put $x = R_K(n)$ and $y = R_K(-n)$. The supporting planes are separated by two, so $(x - y) \cdot n = 2$. The diameter is two, hence $|x - y| \leq 2$. Equality in Cauchy–Schwarz gives $x - y = 2n$. $\square$

Lemma 2.3 (Normal-sphere partition). *For every generic regular-tetrahedron confocal Peabody, the Gauss images of the ten smooth patch interiors together with the four vertex normal cones partition $\mathbb{S}^2$ up to spherical measure zero. Their interiors are pairwise disjoint. The Gauss images of the seams have spherical area zero.*

Proof. A body of constant width is strictly convex. Therefore each $n \in \mathbb{S}^2$ has a unique support point. Two distinct smooth patch interiors cannot have a common outward normal, since the corresponding support plane would touch the body at two points. Every support point lies in a patch interior, a seam, or a vertex, so the listed sets cover $\mathbb{S}^2$.

Each seam is a compact piecewise-smooth curve. Adjacent patches have a common tangent plane along it, so its common normal image is a finite union of curves in $\mathbb{S}^2$. Such a set has two-dimensional spherical measure zero. $\square$

Let Ω_A be the Gauss image of the radius-two cap centered at A , and let $N_K(A)$ be the vertex normal cone.

Lemma 2.4 (Cap–cone duality). *For each tetrahedral vertex,*

$$N_K(A) = -\Omega_A. \quad (2.9)$$

Proof. If $x \in C_A^\circ$, then its outward normal is $n = (x - A)/2$. By Theorem 2.2, $R_K(-n) = A$, so $-\Omega_A \subset N_K(A)$. Conversely, if $m \in N_K(A)$, then $R_K(-m) = A - 2m$ lies on the opposite radius-two cap centered at A , which gives the reverse inclusion. $\square$

For the i -th opposite-edge pair, let $\Gamma_i = \text{Gauss}(W_i^+)$. The binormal pairing of [1, Lemma 3.8] gives

$$\text{Gauss}(W_i^-) = -\Gamma_i.$$

Using Theorems 2.3 and 2.4,

$$4\pi = 2 \sum_{v \in \{A, B, C, D\}} |\Omega_v| + 2 \sum_{i=1}^3 |\Gamma_i|. \quad (2.10)$$

A patch on a sphere of radius two has area four times the area of its Gauss image. Therefore the total cap area is

$$\sum_v |C_v| = 8\pi - 4 \sum_{i=1}^3 |\Gamma_i|. \quad (2.11)$$

Adding the wedge-pod areas gives

$$S_2 = 8\pi + \sum_{i=1}^3 \Psi(e_i), \quad \Psi(e_i) := |W_i^+| + |W_i^-| - 4|\Gamma_i|. \quad (2.12)$$

Interchanging the two devices exchanges W_i^+ and W_i^- , so Ψ is orientation independent.

The generic identity extends to mixed tuples with one or more $e_i = 0$ by the continuous circle-line degeneration in [1, Section 5.3]; equivalently, a collapsed patch may be retained as a zero-area term.

For a body of constant width d , Blaschke's identity is

$$V = \frac{d}{2}S - \frac{\pi}{3}d^3. \quad (2.13)$$

At width two, (2.12) gives

$$V_2 = \frac{16\pi}{3} + \sum_{i=1}^3 \Psi(e_i).$$

Scaling by one half multiplies volume by $1/8$. Define

$$\Phi(e) := \frac{\Psi(e) - \Psi(0)}{8}. \quad (2.14)$$

Since the all-zero body is Meissner, we obtain:

Proposition 2.5 (Exact additive reduction). *For every width-one regular-tetrahedron confocal Peabody,*

$$\text{Vol } K_1(\mathbf{e}, \sigma) = V_M + \Phi(e_1) + \Phi(e_2) + \Phi(e_3), \quad (2.15)$$

independently of σ .

2.3 The eight zero-parameter orientations

Lemma 2.6 (Two equality orbits). *At $\mathbf{e} = 0$, the eight orientation choices form exactly two tetrahedral congruence classes: the four triples of edges incident to a vertex and the four triples bounding a face. These are the two classical Meissner types.*

Proof. Selecting one edge from each of

$$\{AB, CD\}, \quad \{AC, BD\}, \quad \{AD, BC\}$$

gives either a vertex star or a face boundary. There are four of each. The tetrahedral symmetry group acts transitively on vertices and on faces, so there are exactly two orbits. $\square$

3 The one-pair formula

We now derive the formula used by the certificate. Put

$$D_0 = \cosh \xi - e \sin t, \quad s = \sqrt{1 - e^2}.$$

From (2.2)–(2.3),

$$\begin{aligned} \frac{|X - Y|^2}{a^2} &= (\sin t - e \cosh \xi)^2 + (1 - e^2)(\cos^2 t + \sinh^2 \xi) \\ &= \cosh^2 \xi - 2e \sin t \cosh \xi + e^2 \sin^2 t \\ &= (\cosh \xi - e \sin t)^2. \end{aligned}$$

Since $D_0 > 0$ on the parameter rectangle,

$$d := |X - Y| = aD_0. \quad (3.1)$$

Together with (2.4) and (2.5), this proves (2.7).

Let

$$n = \frac{X - Y}{d} = \frac{U}{D_0},$$

where

$$U = (\sin t - e \cosh \xi) \mathbf{k} + s \cos t \mathbf{p} - s \sinh \xi \mathbf{q}.$$

Since $|U| = D_0$,

$$|n_r|^2 = \frac{|U_r|^2 - D_{0,r}^2}{D_0^2}.$$

For $r = t$,

$$U_t = \cos t \mathbf{k} - s \sin t \mathbf{p}, \quad D_{0,t} = -e \cos t,$$

so $|U_t|^2 - D_{0,t}^2 = s^2$. For $r = \xi$,

$$U_\xi = -e \sinh \xi \mathbf{k} - s \cosh \xi \mathbf{q}, \quad D_{0,\xi} = \sinh \xi,$$

and again $|U_\xi|^2 - D_{0,\xi}^2 = s^2$. Finally,

$$U_t \cdot U_\xi = D_{0,t} D_{0,\xi},$$

so

$$n_t \cdot n_\xi = 0, \quad |n_t| = |n_\xi| = \frac{s}{D_0} = \frac{b}{d}. \quad (3.2)$$

Thus

$$d\omega = \frac{b^2}{d^2} dt d\xi. \quad (3.3)$$

The paired boundary points are

$$U_E = X + R_E n, \quad U_H = Y - R_H n.$$

Using $X = Y + dn$ and $d + R_E + R_H = 2$,

$$U_E = Y + (2 - R_H)n, \quad U_H = X - (2 - R_E)n.$$

Because Y, R_H are independent of t , while X, R_E are independent of ξ ,

$$\begin{aligned} (U_E)_t &= (2 - R_H)n_t, & (U_E)_\xi &= R_E n_\xi, \\ (U_H)_t &= -R_H n_t, & (U_H)_\xi &= -(2 - R_E)n_\xi. \end{aligned}$$

Consequently,

$$dA_E = R_E(2 - R_H) \frac{b^2}{d^2} dt d\xi, \quad dA_H = R_H(2 - R_E) \frac{b^2}{d^2} dt d\xi.$$

Since $R_E + R_H = 2 - d$,

$$R_E(2 - R_H) + R_H(2 - R_E) - 4 = -2(d + R_E R_H).$$

Therefore

$$\Psi(e) = -2b^2 \int_\theta^{\pi-\theta} \int_{-\eta}^\eta \frac{d + R_E R_H}{d^2} d\xi dt. \quad (3.4)$$

For fixed t , put $C = e \sin t$ and $y = \tanh(\eta/2)$. The substitution $z = \tanh(\xi/2)$ gives

$$\cosh \xi = \frac{1+z^2}{1-z^2}, \quad d\xi = \frac{2 dz}{1-z^2},$$

and hence

$$I(C) := \int_{-\eta}^{\eta} \frac{d\xi}{\cosh \xi - C} = \frac{4}{\sqrt{1-C^2}} \arctan \left(y \sqrt{\frac{1+C}{1-C}} \right). \quad (3.5)$$

Differentiating this expression and using

$$\frac{2y}{1-y^2} = \sinh \eta = \frac{1}{b}$$

gives

$$\int_{-\eta}^{\eta} \frac{v_0 - \cosh \xi}{(\cosh \xi - C)^2} d\xi = \frac{(v_0 C - 1)I(C) + 2/b}{1 - C^2}. \quad (3.6)$$

Substitution into (3.4) and the symmetry $\sin(\pi - t) = \sin t$ yield

$$\begin{aligned} \Psi(e) = -4b^2 \int_{\theta}^{\pi/2} & \left[\frac{I(e \sin t)}{a} \right. \\ & \left. + \frac{e(\sin t - u_0)}{1 - e^2 \sin^2 t} \left((v_0 e \sin t - 1)I(e \sin t) + \frac{2}{b} \right) \right] dt. \end{aligned} \quad (3.7)$$

At $e = 0$,

$$\Psi(0) = -\frac{2\pi}{\sqrt{3}} \arccos \frac{1}{3}. \quad (3.8)$$

Finally set $x = b \cos t$. Since $b \cos \theta = 1$, this sends $[\theta, \pi/2]$ to $[1, 0]$, and

$$A(e, x) = \sin t = \sqrt{1 - \frac{x^2}{b^2}}, \quad dt = -\frac{dx}{bA}.$$

Thus

$$\Psi(e) = \int_0^1 H(e, x) dx, \quad (3.9)$$

where

$$H(e, x) = -\frac{4b}{A} \left[\frac{I(eA)}{a} + \frac{e(A - u_0)}{1 - e^2 A^2} \left((v_0 e A - 1)I(eA) + \frac{2}{b} \right) \right]. \quad (3.10)$$

The stable parameter

$$q = \frac{1-e}{1+e}, \quad e = \frac{1-q}{1+q} \quad (3.11)$$

fixes the parabolic endpoint at $q = 0$. The complete stable formula, branch control, and endpoint analyticity are proved in Section A.

4 Endpoint value, strict concavity, and scalar positivity

The stable formula extends real analytically to $0 \leq q \leq 1$, and hence Φ extends to $[0, 1]$. At $e = 0$, exact differentiation gives

$$\begin{aligned} \Phi'(0) = \frac{1}{2} \left[\frac{5\pi}{3\sqrt{3}} - 3 \right. \\ \left. + \arccos \frac{1}{3} \left(\frac{3\sqrt{2}}{2} - \frac{5\sqrt{6}\pi}{18} \right) \right] > 0. \end{aligned} \quad (4.1)$$

The endpoint derivative is a consistency check; the compressed proof uses the parabolic endpoint and concavity.

Lemma 4.1 (Parabolic endpoint). *The endpoint extension satisfies*

$$\Phi(0) = 0, \quad \Phi(1) > \frac{1}{4000}. \quad (4.2)$$

Proof. At $q = 0$, put $K = (1 + \sqrt{2})^2$ and $A = 2K + x^2$. The stable integrand simplifies to

$$H(0, x) = \left[-\frac{16\sqrt{2}(1 + \sqrt{2})}{\sqrt{A}} + \frac{4(1 - x^2)(A - 1)}{A^{3/2}} \right] \arctan \frac{1}{\sqrt{A}} - \frac{4(1 - x^2)}{A}. \quad (4.3)$$

The directed-rounding MPFR enclosure described in Section B gives

$$\Phi(1) > 0.0003076528682490342053580258083025 \dots > \frac{1}{4000}.$$

□

Lemma 4.2 (Strict concavity). *For every $0 < e < 1$,*

$$\Phi''(e) < -\frac{1}{30000}. \quad (4.4)$$

Proof. Let $\Psi(q) = \int_0^1 H(q, x) \, dx$. Since

$$q'(e) = -\frac{(1 + q)^2}{2}, \quad q''(e) = \frac{(1 + q)^3}{2},$$

the chain rule gives

$$\Phi''(e) = \frac{(1 + q)^3}{32} \int_0^1 ((1 + q)H_{qq}(q, x) + 2H_q(q, x)) \, dx. \quad (4.5)$$

The direct MPFR certificate of Section B partitions $[0, 1]$ into 32 exact rational slabs and uses ten rational x -panels on each slab. Its weakest upper endpoint is

$$-0.0000416491903361455479929355233265 \dots < -\frac{1}{30000}.$$

□

Lemma 4.3 (Chord bound). *For every $0 < e < 1$,*

$$\Phi(e) > \frac{e}{4000} > 0. \quad (4.6)$$

Proof. By Theorem 4.2, Φ is strictly concave. Therefore it lies above the chord joining its endpoint values:

$$\Phi(e) \geq (1 - e)\Phi(0) + e\Phi(1).$$

Apply Theorem 4.1. □

5 Proof of the main theorem

Proof of Theorem 1.1. By dilation it is enough to take $d = 1$. From Theorems 2.5 and 4.3,

$$\text{Vol } K_1(\mathbf{e}, \sigma) - V_M = \Phi(e_1) + \Phi(e_2) + \Phi(e_3) \geq 0.$$

Equality forces $e_1 = e_2 = e_3 = 0$, and the converse is immediate. The two equality congruence classes are identified by Theorem 2.6. Scaling by d proves (1.2). $\square$

6 Scope and reproducibility

The theorem concerns only regular-tetrahedron confocal Peabodies. It does not prove the global Meissner conjecture, improve a universal lower bound, or establish a six-patch construction for arbitrary semi-regular tetrahedral bases.

The exact part of the proof consists of the source construction, Theorems 2.1, 2.3, 2.4 and 2.6, the width normalization, and the one-pair formulas. The finite proof authority is an independently written C implementation using MPFR directed rounding. Lower endpoints are evaluated with MPFR_RNDD, upper endpoints with MPFR_RNDU; this includes direct calls to `mpfr_sqrt` and `mpfr_atan`. The canonical proof uses four endpoint panels and 32 parameter slabs with ten integration panels each. It is replayed at 384 and 512 bits and in reverse slab order. A 16-slab under-resolved control and a correction-sector sign mutation are rejected. The earlier `mpmath.iv`, Mathematica, and R calculations are retained only as redundant audits.

The public major-revision checkpoint includes the C source, exact rational partitions, all output endpoints, formula-identity checks, proof notes, and replay scripts. Before submission, the official-header MPFR replay and independent geometric/formula proofreading are required.

A Stable chart, branch control, and parabolic analyticity

Put

$$\kappa = 1 + \sqrt{2}, \quad K = \kappa^2 = 3 + 2\sqrt{2}.$$

From (3.11),

$$1 - e^2 = \frac{4q}{(1+q)^2}, \quad b^2 = \frac{K^2 + q^2}{2Kq}, \quad a^2 = \frac{(K^2 + q^2)(1+q)^2}{8Kq^2}.$$

Define

$$\begin{aligned} D &= \sqrt{K^2 + q^2}, \\ R &= \sqrt{K^2 + q^2 - 2Kqx^2}, \\ T &= \sqrt{2K(K^2 + q^2) + K^2(1-q)^2x^2}. \end{aligned}$$

The apparent difference $(R - K + q)/q$ is rationalized using

$$R^2 - (K - q)^2 = 2Kq(1 - x^2),$$

which gives

$$\delta = \frac{2K(1 - x^2)}{R + K - q}.$$

Likewise,

$$R - K = \frac{q(q - 2Kx^2)}{R + K}$$

reduces the second removable quotient to

$$M = \frac{K(q - 2Kx^2)}{R + K} + (1 - K)R - K^2 - qR - q - q^2.$$

Set

$$\begin{aligned} P_1 &= -\frac{16\sqrt{2}\kappa(K^2 + q^2)}{RT}, \\ P_2 &= -\frac{4K(1 - q^2)(K^2 + q^2)\delta M}{RT^3}, \\ Q_0 &= -\frac{4K(1 - q^2)(K^2 + q^2)\delta}{RT^2}, \\ Z &= \frac{K((1 + q)D + (1 - q)R)}{T(K + q + D)}. \end{aligned}$$

Then

$$H(q, x) = (P_1 + P_2) \arctan Z + Q_0. \quad (\text{A.1})$$

We now fix the branch. In the original formula,

$$C = e \sin t \in [0, 1), \quad y = \tanh(\eta/2) > 0,$$

so the argument in (3.5) is positive and represents an angle in $[0, \pi/2)$. Put

$$A_+ = (1 + q)D + (1 - q)R, \quad A_- = (1 + q)D - (1 - q)R.$$

A direct calculation gives

$$A_+ A_- = \frac{2q}{K} T^2, \quad K + q - D = \frac{2Kq}{K + q + D}.$$

Using these positive identities gives, without squaring or changing signs,

$$y \sqrt{\frac{1 + C}{1 - C}} = \frac{K A_+}{T(K + q + D)} = Z.$$

Therefore the principal $\arctan Z$ in (A.1) is the same analytic branch as the geometric angle on the entire domain.

Finally, on $0 \leq q, x \leq 1$,

$$D^2 \geq K^2, \quad R^2 \geq (K - q)^2 \geq (K - 1)^2 > 0, \quad T^2 \geq 2K^3 > 0.$$

Also

$$R + K - q \geq 2(K - 1) > 0, \quad R + K > 0, \quad K + q + D > 0.$$

Thus every radicand and denominator is uniformly separated from zero; $Z > 0$; and (A.1) contains no negative power of q . Hence H is real analytic on a neighborhood of the closed square $[0, 1]^2$. Differentiation under the fixed integral is justified, including on slabs meeting $q = 0$.

B Direct MPFR verification

For an interval $[a, b]$, the certifier implements elementary endpoint arithmetic with directed MPFR rounding. For example,

$$[a, b] + [c, d] = [\text{RNDD}(a + c), \text{RNDU}(b + d)],$$

and multiplication takes the minimum of the four downward-rounded endpoint products and the maximum of the four upward-rounded products. On positive intervals,

$$\sqrt{[a, b]} = [\text{RNDD}(\sqrt{a}), \text{RNDU}(\sqrt{b})],$$

while monotonicity gives

$$\arctan([a, b]) = [\text{RNDD}(\arctan a), \text{RNDU}(\arctan b)].$$

The implementation calls `mpfr_sqrt` and `mpfr_atan` directly.

For an x -panel $[a, b]$, with midpoint m and width h , the certifier uses the rigorous midpoint enclosure

$$\int_a^b f(x) \, dx \in hf(m) + \frac{h^3}{24} f''([a, b]). \quad (\text{B.1})$$

For the endpoint lemma, four equal rational panels enclose $\int_0^1 H(0, x) \, dx$.

For concavity, put

$$C(q, x) = (1 + q)H_{q,q}(q, x) + 2H_q(q, x).$$

A nested automatic-differentiation jet encloses the derivatives through third order in q and second order in x . The interval $[0, 1]$ is partitioned into

$$Q_j = \left[\frac{j}{32}, \frac{j+1}{32} \right], \quad j = 0, \dots, 31.$$

On each slab, centered at q_j , the mean-value form

$$\int_0^1 C(Q_j, x) \, dx \subseteq \int_0^1 C(q_j, x) \, dx + (Q_j - q_j) \int_0^1 C_q(Q_j, x) \, dx$$

is combined with ten rational x -panels and (B.1). Multiplication by $(1 + Q_j)^3/32$ encloses Φ'' .

The 384-bit forward certificate gives

$$\Phi(1) > 0.0003076528682490342053580258083025767 \dots > \frac{1}{4000},$$

and

$$\sup_{0 < e < 1} \Phi''(e) < -0.0000416491903361455479929355233265745 \dots < -\frac{1}{30000}.$$

The 512-bit and reverse-order replays give the same reported enclosures. The 16-slab control and the sign mutation fail, as required. All 32 MPFR slab enclosures overlap the earlier independent `mpmath.iv` enclosures.

References

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